Canto-L · training sample

Canto-L and acorn trained on the HNL signal, on ttbar + HNL and on ttbar, all with the same track builder · HNL validation and one-time test

	Canto-L			acorn		
	Canto-L (HNL) 1 seed · HNL signal learned merging τ 0.2	Canto-L (ttbar+HNL) 1 seed · ttbar+HNL learned merging τ 0.3	Canto-L (ttbar) 3 seeds · ttbar learned merging	acorn (HNL) 3 seeds · HNL signal learned merging	acorn (ttbar+HNL) 3 seeds · ttbar+HNL learned merging	acorn (ttbar) 3 seeds · ttbar learned merging
HNL samples · validation pooled HNL val (hnl10_100, hnl15_200; 1,999 events) and ttbar val · mean over the seeds in the header · every model at a track fake rate ≤ FR* = 2.94 %, its builder setting chosen per seed on these events						
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm · 1,331 per seed	99.17% ± 0.32	**99.25%** ± 0.31	99.10% ± 0.27	98.65% ± 0.31	97.90% ± 0.40	97.17% ± 0.44
Far-displaced tracks |d0| 100–300 mm · 167 per seed	**98.80%** 165/167	**98.80%** 165/167	97.60% ± 1.28	97.21% ± 1.32	95.61% ± 1.70	93.81% ± 2.22
Very far-displaced |d0| ≥ 300 mm · k/n · 19 per seed	**94.7%** 18/19	**94.7%** 18/19	87.7% 50/57	91.2% 52/57	84.2% 48/57	77.2% 44/57
Prompt muons W → μ in HNL events	**99.79%** 1865/1869	**99.79%** 1865/1869	**99.79%** ± 0.15	99.75% ± 0.16	99.75% ± 0.15	**99.79%** ± 0.15
Dense prompt tracks ttbar val, |d0| < 1 mm	**96.69%** 50957/52702	96.41% 50809/52702	96.31% ± 0.25	94.91% ± 0.16	95.18% ± 0.35	95.49% ± 0.24
Fake-track rate HNL events · lower is better	1.02% 1 seed, no σ	**1.01%** 1 seed, no σ	1.03% ± 0.05	1.37% ± 0.06	1.39% ± 0.07	1.53% ± 0.06
HNL samples · one-time test pooled HNL test (hnl10_100_test, hnl15_200_test, hnl15_100_external; 6,999 events) and ttbar_test · evaluated once at the settings chosen on val · Canto-L (HNL) and Canto-L (ttbar+HNL): trained after the freeze, settings committed before their test jobs (post-freeze)						
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm · 5,013 per seed	98.94% ± 0.24	98.96% ± 0.23	**99.08%** ± 0.15	97.98% ± 0.20	97.37% ± 0.31	96.28% ± 0.56
Far-displaced tracks |d0| 100–300 mm · 584 per seed	**98.97%** ± 0.50	98.80% ± 0.54	98.57% ± 0.39	96.75% ± 0.59	95.49% ± 1.01	91.84% ± 1.80
Very far-displaced |d0| ≥ 300 mm · k/n · 36 per seed	91.7% 33/36	91.7% 33/36	87.0% 94/108	**93.5%** 101/108	90.7% 98/108	88.9% 96/108
Prompt muons W → μ in HNL events	99.74% ± 0.07	99.75% ± 0.06	**99.77%** ± 0.06	99.69% ± 0.07	99.65% ± 0.07	99.68% ± 0.08
Dense prompt tracks ttbar_test, |d0| < 1 mm	**96.53%** ± 0.36	96.35% ± 0.37	96.32% ± 0.24	94.84% ± 0.20	95.17% ± 0.34	95.40% ± 0.22
Fake-track rate HNL events · lower is better	1.01% ± 0.04	**1.00%** ± 0.04	1.02% ± 0.03	1.36% ± 0.03	1.38% ± 0.05	1.52% ± 0.03
Displaced tracks hnl10_100_test only · 603 per seed	98.84% ± 0.46	99.00% ± 0.42	**99.23%** ± 0.33	98.45% ± 0.47	97.51% ± 0.75	97.24% ± 0.65
Displaced tracks hnl15_200_test only · 549 per seed	**99.45%** ± 0.33	98.36% ± 0.62	98.48% ± 0.47	97.02% ± 0.70	96.17% ± 0.90	94.54% ± 1.02
Displaced tracks hnl15_100_external only · 3,861 per seed · its mass and lifetime in no training sample	98.89% ± 0.28	99.04% ± 0.27	**99.14%** ± 0.17	98.04% ± 0.22	97.51% ± 0.28	96.37% ± 0.61
Training · edge classifier the HNL-trained and ttbar+HNL-trained models use the training split of hnl10_100 and hnl15_200						
Training data events, one graph per event	HNL signal 15,999 events	ttbar + HNL 15,998 events	ttbar 7,999 events	HNL signal 15,999 events	ttbar + HNL 15,998 events	ttbar 7,999 events
Parameters edge classifier	347,777	347,777	347,777	263,937	263,937	263,937
Training steps one graph per step	479,970 30 epochs	479,940 30 epochs	239,970 30 epochs	479,970 30 epochs	479,940 30 epochs	239,970 30 epochs

Samples and definitions.

ACTS generic detector simulation without pileup: HNL signal samples (m_N, cτ in the sample name) and ttbar. Track efficiency by double-majority matching: more than half of the track's hits come from the particle and more than half of the particle's hits are on the track. Targets: pT > 1 GeV, |η| < 4, at least 5 hits. Displaced tracks: HNL-daughter targets with |d0| ≥ 10 mm or r_v ≥ 100 mm. Fake-track rate: the fraction of tracks with at least 5 hits that match no particle.

Models and working points.

Every column runs MWS-layer + learned merging, the frozen track builder of Canto-L, at a track fake rate ≤ FR* = 2.94 % on pooled HNL val (preregistered). Canto-L (ttbar): the frozen Canto-L, selected by rules R6 and R8 before the test (results/final_selection_v1.json). Canto-L (HNL) and Canto-L (ttbar+HNL): the same recipe with only the training sample changed, one seed each, the builder setting (header) chosen on val by the builder rule at FR*; trained after the freeze, evaluated once on test (post-freeze, descriptive). acorn (HNL), acorn (ttbar+HNL) and acorn (ttbar): acorn's interaction network trained on the same three samples, three seeds each, with the val-best builder of the full menu chosen per seed (MWS-layer + learned merging for every seed).

Seeds and uncertainties.

Means over the training seeds in the column header; k/n pooled over the seeds. Second line of a cell: the combined uncertainty σ = (σ_boot² + s²/n)^½ of the mean, in percentage points, with σ_boot from an event bootstrap (B = 1,000 on val, 500 on test) and s the standard deviation over the n seeds. The |d0| ≥ 300 mm rows show k/n instead (statistics limited). Canto-L (HNL) and Canto-L (ttbar+HNL): one seed; their σ takes the seed standard deviation of Canto-L (ttbar) in the same row as its seed term; on val only the displaced-track row has a σ in the source, the other val rows show k/n, or '1 seed, no σ' for the fake rate.

Shading.

Framed in orange: the first place of each row, the highest value (the lowest for rates; cells showing the same value share it), in whichever column it is. A difference below 2σ, σ the two cells' uncertainties in quadrature, is not resolved. Paired differences on the same events, with smaller uncertainties: results/training_sample_paired_test_v1.csv (test) and results/training_sample_v1.md (val).

Training.

Edge classifier only (results/model_table_v1.md); the metric-learning graph stage is not included. One step is one event graph. hnl15_100_external (m_N = 15 GeV, cτ = 100 mm) is Wang et al.'s evaluation sample and was used by no model in training or selection. All numbers of this card: HNL samples only; Canto-L on TrackML is on its own card.